

Complete System Design Learning Roadmap

Beginner → Intermediate → FAANG-Level Preparation

This roadmap is designed to help you learn:

- OOP & Clean Code
- SOLID Principles
- Design Patterns
- Low Level Design (LLD)
- High Level System Design (HLD)
- Distributed Systems
- Real-World Architecture

PHASE 1 — OOP & Basics

Recommended Duration: 1–2 Weeks

Channel	Description	Playlist	Direct Link
CodeWithHarry	Python & OOP Basics	Python Tutorials	https://www.youtube.com/@CodeWithHarry
Apna College	Java + OOP Fundamentals	Java Placement Course	https://www.youtube.com/@ApnaCollegeOfficial

PHASE 2 — SOLID Principles & Design Patterns

Recommended Duration: 2–3 Weeks

Channel	Description	Playlist	Direct Link
Concept & Coding	Best Hindi explanation for SOLID + LLD	LLD Playlist	https://www.youtube.com/@ConceptAndCoding
freeCodeCamp	Deep clean-code explanations	SOLID Principles Course	https://www.youtube.com/@freecodecamp

PHASE 3 — Low Level Design

Recommended Duration: 3–5 Weeks

Channel	Description	Playlist	Direct Link
Code Decode	LLD Interview Preparation	LLD Playlist	https://www.youtube.com/@CodeDecode
Concept & Coding	Factory, Singleton, Observer etc.	Design Patterns	https://www.youtube.com/@ConceptAndCoding

PHASE 4 — System Design Basics

Recommended Duration: 3–4 Weeks

Channel	Description	Playlist	Direct Link
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Gaurav Sen	Best beginner-friendly System Design	System Design Playlist	https://www.youtube.com/@gkcs
ByteByteGo	Industry-level architecture	System Design Interview	https://www.youtube.com/@ByteByteGo

PHASE 5 — Advanced System Design

Recommended Duration: 4–8 Weeks

Channel	Description	Playlist	Direct Link
Hello Interview	FAANG mock interviews	System Design Questions	https://www.youtube.com/@hello_interview
ByteByteGo	Distributed Systems	Advanced System Design	https://bytebytego.com/courses/system-design

PHASE 6 — Real Engineering

Recommended Duration: Ongoing

Channel	Description	Playlist	Direct Link
Hussein Nasser	Backend & distributed systems	Backend Engineering	https://www.youtube.com/@hnasr
NeetCode	Problem-solving mindset	System Design + DSA	https://www.youtube.com/@NeetCode

Recommended Order:

1. Python / Java OOP Basics
2. SOLID Principles
3. Design Patterns
4. Low Level Design (LLD)
5. Databases + APIs + Caching
6. System Design Basics
7. Distributed Systems
8. Build Real Projects
9. FAANG-Level Interview Preparation